



1. Description

1.1. Project

Project Name	ORION V2
Board Name	NUCLEO-H755ZI-Q
Generated with:	STM32CubeMX 6.18.1
Date	09/02/2026

1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H745/755
MCU name	STM32H755ZITx
MCU Package	LQFP144
MCU Pin number	144

1.3. Core(s) information

Core(s)	ARM Cortex-M7 ARM Cortex-M4
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2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
6	VSS	Power		
7	VDD	Power		
8	VBAT	Power		
9	PC13	I/O	GPIO_EXTI13	
10	PC14-OSC32_IN (OSC32_IN) *	I/O	RCC_OSC32_IN	
11	PC15-OSC32_OUT (OSC32_OUT) *	I/O	RCC_OSC32_OUT	
12	VSS	Power		
13	VDD	Power		
14	VSSMPS	Power		
15	VLXSMPS	Power		
16	VDDSMPS	Power		
17	VFBSMPS	Power		
18	VSS	Power		
19	VDD	Power		
25	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
26	PH1-OSC_OUT (PH1) *	I/O	RCC_OSC_OUT	
27	NRST	Reset		
29	PC1	I/O	ETH_MDC	
32	VDD	Power		
33	VSS	Power		
34	VSSA	Power		
36	VDDA	Power		
38	PA1	I/O	ETH_REF_CLK	
39	PA2	I/O	ETH_MDIO	
41	VSS	Power		
42	VDD	Power		
44	PA5	I/O	SPI1_SCK	
45	PA6	I/O	TIM3_CH1	
46	PA7	I/O	ETH_CRS_DV	
47	PC4	I/O	ETH_RXD0	
48	PC5	I/O	ETH_RXD1	
55	VSS	Power		
56	VDD	Power		
58	PE8	I/O	GPIO_EXTI8	VL_INT

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
59	PE9	I/O	TIM1_CH1	
61	PE11	I/O	TIM1_CH2	
66	PB10	I/O	I2C2_SCL	
67	PB11	I/O	I2C2_SDA	
68	VCAP	Power		
69	VSS	Power		
70	VDDLDO	Power		
71	VDD	Power		
73	PB13	I/O	ETH_TXD1	
74	PB14 **	I/O	GPIO_Output	LD3
79	VDD	Power		
80	VSS	Power		
82	PD12	I/O	TIM4_CH1	
83	PD13	I/O	TIM4_CH2	
89	VSS	Power		
90	VDD50_USB	Power		
91	VDD33_USB	Power		
92	VDD	Power		
93	PC6 **	I/O	GPIO_Output	OLED_RST
94	PC7	I/O	TIM8_CH2	
103	VCAP	Power		
104	VSS	Power		
105	VDDLDO	Power		
106	VDD	Power		
113	PD1 **	I/O	GPIO_Output	M1DIR
114	PD2 **	I/O	GPIO_Output	M2DIR
118	VSS	Power		
119	VDD	Power		
120	PD6 **	I/O	GPIO_Output	BUZZ
121	PD7	I/O	SPI1_MOSI	
124	PG11	I/O	ETH_TX_EN	
126	PG13	I/O	ETH_TXD0	
128	VSS	Power		
129	VDD	Power		
132	PB5	I/O	TIM3_CH2	
133	PB6	I/O	I2C1_SCL	
134	PB7	I/O	I2C1_SDA	
135	BOOT0	Boot		
136	PB8 **	I/O	GPIO_Output	OLED_CS

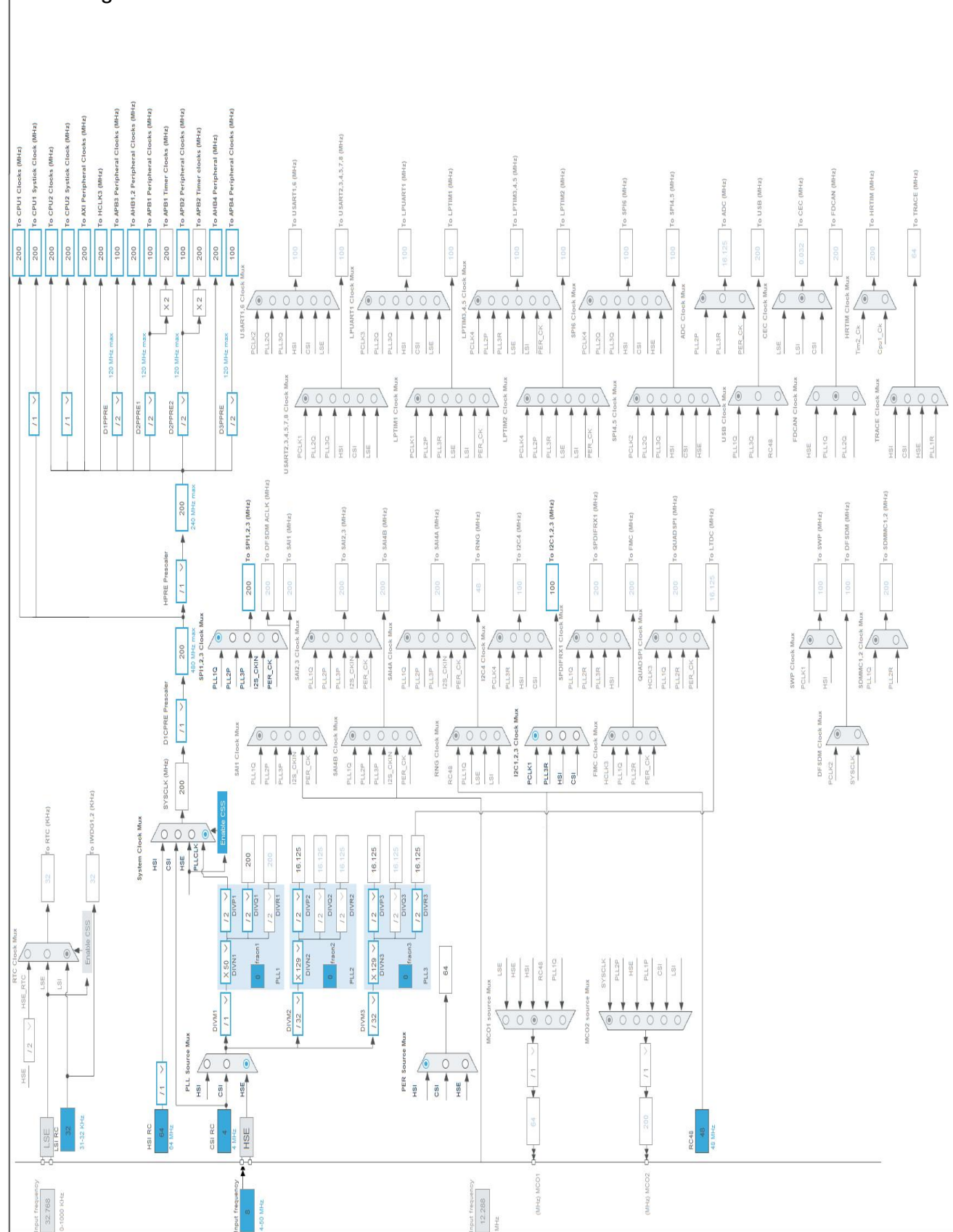
Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
137	PB9 **	I/O	GPIO_Output	OLED_DC
139	PE1 **	I/O	GPIO_Output	LD2
140	VCAP	Power		
141	VSS	Power		
142	PDR_ON	Power		
143	VDDLDO	Power		
144	VDD	Power		

** The pin is affected with an I/O function

* The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration

RIF configuration not available for STM32H755ZITx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H745/755
MCU	STM32H755ZITx
Datasheet	DS12919 Rev1

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(DD36000)
Capacity	36000.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	450.0 mA
Max Pulse Current	1000.0 mA
Cells in series	1
Cells in parallel	1

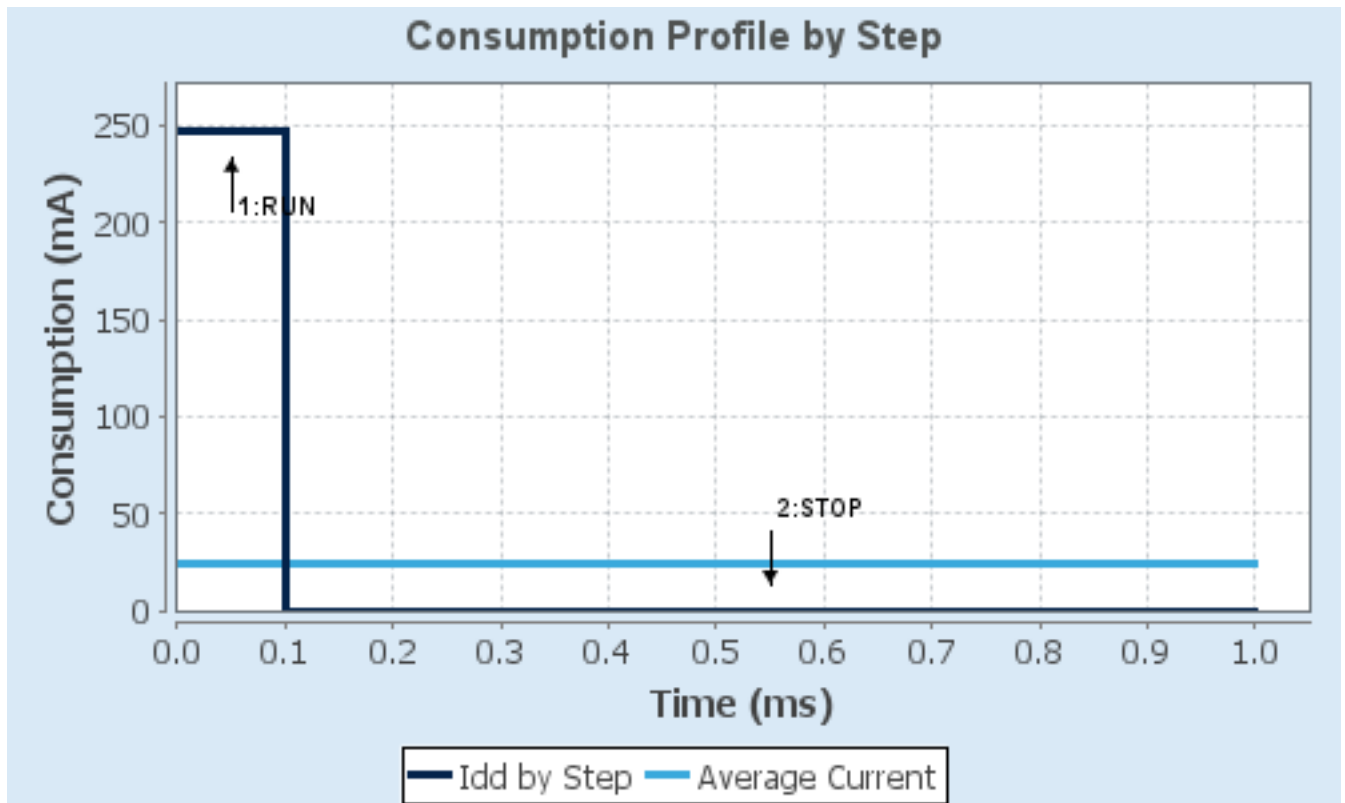
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0	SVOS5: System-Scale5
D1 Mode	DRUN/CRUN	DSTANDBY
D2 Mode	DRUN/CRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	CM7: ITCM/Cache / CM4: FLASH_B/ART	CM7: NA / CM4: NA
CM7 Frequency	480 MHz	0 Hz
Clock Configuration	HSE BYP PLL ALL_IPs_ON	LSE Flash-ON
CM4 Frequency	240 MHz	0 Hz
Clock Source Frequency	25 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	247 mA	145 μ A
Duration	0.1 ms	0.9 ms
DMIPS	1027.0	0.0
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	24.83 mA
Battery Life	1 month, 29 days, 21 hours	Average DMIPS	1027.2001 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	ORION_V2
Project Folder	C:\Users\Oleg\STM32CubeIDE\workspace_2.1.1\ORION_V2
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_H7 V1.13.0
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	M4-0x200
Minimum Stack Size	M4-0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls ARM Cortex-M7

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_LWIP_Init	LWIP
5	MX_SPI1_Init	SPI1

2.4. Advanced Settings - Generated Function Calls ARM Cortex-M4

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO

Rank	Function Name	Peripheral Instance Name
2	MX_DMA_Init	DMA
3	MX_TIM4_Init	TIM4
4	MX_TIM3_Init	TIM3
5	MX_I2C1_Init	I2C1
6	MX_I2C2_Init	I2C2
7	MX_TIM1_Init	TIM1
8	MX_TIM8_Init	TIM8

3. Peripherals and Middlewares Configuration

3.1. CORTEX_M7

3.1.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Speculation default mode Settings:

Speculation default mode	Disabled
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Cortex Interface Settings:

CPU ICache	Enabled *
CPU DCache	Enabled *

Cortex Memory Protection Unit Control Settings:

MPU Control Mode	Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers *
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Cortex Memory Protection Unit Region 0 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x24000000 *
MPU Region Size	512KB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 1 *
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	ENABLE
MPU Shareability Permission	ENABLE *
MPU Cacheable Permission	ENABLE *
MPU Bufferable Permission	ENABLE *

Cortex Memory Protection Unit Region 1 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x30040000 *
MPU Region Size	32KB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 1 *
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	ENABLE
MPU Shareability Permission	

ENABLE *

MPU Cacheable Permission DISABLE

MPU Bufferable Permission DISABLE

Cortex Memory Protection Unit Region 2 Settings:

MPU Region **Enabled ***

MPU Region Base Address **0x38000000 ***

MPU Region Size **64KB ***

MPU SubRegion Disable **0x0 ***

MPU TEX field level level 0

MPU Access Permission **ALL ACCESS PERMITTED ***

MPU Instruction Access **DISABLE ***

MPU Shareability Permission **ENABLE ***

MPU Cacheable Permission DISABLE

MPU Bufferable Permission DISABLE

Cortex Memory Protection Unit Region 3 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 4 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 5 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 6 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 7 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 8 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 9 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 10 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 11 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 12 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 13 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region Disabled

3.2. ETH

Mode: RMII

3.2.1. Parameter Settings:

Core(s) Settings:

Context(s): Cortex-M7
 Initialized Context: Cortex-M7
 Power Domain: D2

General : Ethernet Configuration:

Warning Ethernet RX and Tx descriptors needs to be placed in a RAM memory accessible by the dedicated Ethernet DMA

Note PHY Driver must be configured from the LwIP 'Platform Settings' top right tab

Ethernet MAC Address 00:80:E1:00:00:00

Tx Descriptor Length 4

First Tx Descriptor Address **0x30000080 ***

Rx Descriptor Length 4

First Rx Descriptor Address **0x30000000 ***

Rx Buffers Address **0x30000100 ***

Rx Buffers Length 1536

3.3. I2C1

I2C: I2C

3.3.1. Parameter Settings:

Core(s) Settings:

Context(s): Cortex-M4
 Initialized Context: Cortex-M4
 Power Domain: D2

Timing configuration:

Custom Timing Disabled

I2C Speed Mode Standard Mode

I2C Speed Frequency (KHz) 100

Rise Time (ns) 0

Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x10C0ECFF *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.4. I2C2

I2C: I2C

3.4.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x10C0ECFF *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.5. RCC

High Speed Clock (HSE): BYPASS Clock Source

3.5.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7 Cortex-M4
Initialized Context:	Cortex-M7
Power Domain:	D3

Power Parameters:

SupplySource	PWR_DIRECT_SMPS_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 1 *

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
CSI Calibration Value	32
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	2 WS (3 CPU cycle)
Product revision	rev.V

PLL range Parameters:

PLL1 clock Input range	Between 8 and 16 MHz
PLL1 clock Output range	Wide VCO range

3.6. SPI1

Mode: Transmit Only Master

3.6.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits *

First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	16 *
Baud Rate	12.5 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.7. SYS

Timebase Source: TIM6

3.7.1. Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	

3.8. SYS_M4

Timebase Source: TIM7

3.8.1. Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	

3.9. TIM1

Combined Channels: Encoder Mode

3.9.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	Encoder Mode TI1 and TI2 *
____ Parameters for Channel 1 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0
____ Parameters for Channel 2 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

3.10. TIM3

Combined Channels: Encoder Mode

3.10.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	Encoder Mode TI1 and TI2 *
____ Parameters for Channel 1 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0
____ Parameters for Channel 2 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

3.11. TIM4

Channel1: PWM Generation CH1

Channel2: PWM Generation CH2

3.11.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Counter Settings:

Prescaler (PSC - 16 bits value)	9 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	624 *
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.12. TIM8

Channel2: PWM Generation CH2

3.12.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	249 *
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0

auto-reload preload	Disable
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Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)

Disable (Trigger input effect not delayed)

Trigger Event Selection TRGO Reset (UG bit from TIMx_EGR)

Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)
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Break And Dead Time management - BRK Configuration:

BRK State Disable

BRK Polarity High

BRK Filter (4 bits value)	0
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BRK Sources Configuration

- Digital Input Disable

- COMP1 Disable

- COMP2 Disable

- DFSDM Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State Disable

BRK2 Polarity High

BRK2 Filter (4 bits value)	0
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BRK2 Sources Configuration

- Digital Input Disable

- COMP1 Disable

- COMP2 Disable

- DFSDM Disable

Break And Dead Time management - Output Configuration:

Automatic Output State Disable

Off State Selection for Run Mode (OSSR)	Disable
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Off State Selection for Idle Mode (OSSI)	Disable
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Lock Configuration	Off
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Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 2:

Mode PWM mode 1

Pulse (16 bits value)	0
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Output compare preload	Enable
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Fast Mode Disable

CH Polarity High

CH Idle StateReset

3.13. ARM.CMSIS.5.7.0_M4

mode: CMSISJjDSP

3.13.1. Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

3.14. FREERTOS_M4

Interface: CMSIS_V2

3.14.1. Config parameters:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

API:

FreeRTOS API	CMSIS v2
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Versions:

FreeRTOS version	10.6.2
CMSIS-RTOS version	2.1.3

MPU/FPU:

ENABLE_MPU	Disabled
ENABLE_FPU	Disabled

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemD2Clock
TICK_RATE_HZ	1000
MAX_PRIORITIES	56
MINIMAL_STACK_SIZE	128
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled
USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled

ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled
Memory management settings:	
Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	32768 *
HEAP_CLEAR_MEMORY_ON_FREE	0
Memory Management scheme	heap_4
Hook function related definitions:	
USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled
Run time and task stats gathering related definitions:	
GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled
Co-routine related definitions:	
USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2
Software timer definitions:	
USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	256
Interrupt nesting behaviour configuration:	
LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5
Added with 10.2.1 support:	
MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled
CMSIS-RTOS V2 flags:	
USE_OS2_THREAD_SUSPEND_RESUME	Enabled
USE_OS2_THREAD_ENUMERATE	Enabled
USE_OS2_EVENTFLAGS_FROM_ISR	Enabled
USE_OS2_THREAD_FLAGS	Enabled
USE_OS2_TIMER	Enabled
USE_OS2_MUTEX	Enabled

3.14.2. Include parameters:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Enabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.14.3. Advanced settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Enabled *
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Project settings (see parameter description first):

Use FW pack heap file

Enabled

3.15. FREERTOS_M7

Interface: CMSIS_V2

3.15.1. Config parameters:

Core(s) Settings:

Context(s):

Cortex-M7

Initialized Context:

Cortex-M7

Power Domain:

D1

API:

FreeRTOS API

CMSIS v2

Versions:

FreeRTOS version

10.6.2

CMSIS-RTOS version

2.1.3

MPU/FPU:

ENABLE_MPU

Disabled

ENABLE_FPU

Disabled

Kernel settings:

USE_PREEMPTION

Enabled

CPU_CLOCK_HZ

SystemCoreClock

TICK_RATE_HZ

1000

MAX_PRIORITIES

56

MINIMAL_STACK_SIZE

128

MAX_TASK_NAME_LEN

16

USE_16_BIT_TICKS

Disabled

IDLE_SHOULD_YIELD

Enabled

USE_MUTEXES

Enabled

USE_RECURSIVE_MUTEXES

Enabled

USE_COUNTING_SEMAPHORES

Enabled

QUEUE_REGISTRY_SIZE

8

USE_APPLICATION_TASK_TAG

Disabled

ENABLE_BACKWARD_COMPATIBILITY

Enabled

USE_PORT_OPTIMISED_TASK_SELECTION

Disabled

USE_TICKLESS_IDLE

Disabled

USE_TASK_NOTIFICATIONS

Enabled

RECORD_STACK_HIGH_ADDRESS

Disabled

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	32767 *
HEAP_CLEAR_MEMORY_ON_FREE	0
Memory Management scheme	heap_4
Hook function related definitions:	
USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled
Run time and task stats gathering related definitions:	
GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled
Co-routine related definitions:	
USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2
Software timer definitions:	
USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	256
Interrupt nesting behaviour configuration:	
LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5
Added with 10.2.1 support:	
MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled
CMSIS-RTOS V2 flags:	
USE_OS2_THREAD_SUSPEND_RESUME	Enabled
USE_OS2_THREAD_ENUMERATE	Enabled
USE_OS2_EVENTFLAGS_FROM_ISR	Enabled
USE_OS2_THREAD_FLAGS	Enabled
USE_OS2_TIMER	Enabled
USE_OS2_MUTEX	Enabled

3.15.2. Include parameters:

Core(s) Settings:

Context(s):	Cortex-M7
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Initialized Context:	Cortex-M7
Power Domain:	D1
Include definitions:	
vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Enabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.15.3. Advanced settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Enabled *
----------------------	------------------

Project settings (see parameter description first):

Use FW pack heap file	Enabled
-----------------------	---------

3.16. LWIP

mode: Enabled

Advanced parameters are not listed except if modified by user.

3.16.1. General Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

LwIP Version:

LwIP Version (Version of LwIP supported by CubeMX ** CubeMX specific **)	2.2.1
--	-------

IPv4 - DHCP Options:

LWIP_DHCP (DHCP Module)	Disabled *
-------------------------	------------

IP Address Settings:

IP_ADDRESS (IP Address)	192.168.001.021 *
NETMASK_ADDRESS (Netmask Address)	255.255.255.000 *
GATEWAY_ADDRESS (Gateway Address)	000.000.000.000

RTOS Dependency:

WITH_RTOS (Use FREERTOS ** CubeMX specific **)	Enabled
CMSIS_VERSION (CMSIS API Version used)	CMSIS v2
RTOS_USE_NEWLIB_REENTRANT (RTOS used - 1)	Enabled

Platform Settings:

PHY Driver	Choose/LAN8742
------------	----------------

Protocols Options:

LWIP_ICMP (ICMP Module Activation)	Enabled
LWIP_IGMP (IGMP Module)	Disabled
LWIP_DNS (DNS Module)	Disabled
LWIP_UDP (UDP Module)	Enabled
MEMP_NUM_UDP_PCB (Number of UDP Connections)	15 *
LWIP_TCP (TCP Module)	Enabled
MEMP_NUM_TCP_PCB (Number of TCP Connections)	5

3.16.2. Key Options:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Infrastructure - OS Awareness Option:

NO_SYS (OS Awareness) OS Used

Infrastructure - Timers Options:

LWIP_TIMERS (Use Support For sys_timeout) Enabled

Infrastructure - Core Locking and MPU Options:

SYS_LIGHTWEIGHT_PROT (Memory Functions Protection) Enabled

Infrastructure - Heap and Memory Pools Options:

MEM_SIZE (Heap Memory Size) **2048 ***
 LWIP_RAM_HEAP_POINTER (RAM Heap Pointer) **0x30046000 ***
 ETH_RX_BUFFER_CNT 12

Infrastructure - Internal Memory Pool Sizes:

MEMP_NUM_PBUF (Number of Memory Pool struct Pbufs) 16
 MEMP_NUM_RAW_PCB (Number of Raw Protocol Control Blocks) 4
 MEMP_NUM_TCP_PCB_LISTEN (Number of Listening TCP Connections) 8
 MEMP_NUM_TCP_SEG (Number of TCP Segments simultaneously queued) 16
 MEMP_NUM_LOCALHOSTLIST (Number of Host Entries in the Local Host List) 1

Pbuf Options:

PBUF_POOL_SIZE (Number of Buffers in the Pbuf Pool) 16
 PBUF_POOL_BUFSIZE (Size of each pbuf in the pbuf pool) 592

IPv4 - ARP Options:

LWIP_ARP (ARP Functionality) Enabled

Callback - TCP Options:

TCP_TTL (Number of Time-To-Live Used by TCP Packets) 255
 TCP_WND (TCP Receive Window Maximum Size) 2144
 TCP_QUEUE_OOSEQ (Allow Out-Of-Order Incoming Packets) Enabled
 LWIP_TCP_SACK_OUT (Allow Sending Selective Acknowledgements) Disabled
 TCP_MSS (Maximum Segment Size) 536
 TCP_SND_BUF (TCP Sender Buffer Space) 1072
 TCP_SND_QUEUELEN (Number of Packet Buffers Allowed for TCP Sender) 9

Network Interfaces Options:

LWIP_NETIF_STATUS_CALLBACK (Callback Function on Interface Status Changes) Disabled
 LWIP_NETIF_EXT_STATUS_CALLBACK (Extended Callback Function for several netif) Disabled
 LWIP_NETIF_LINK_CALLBACK (Callback Function on Interface Link Changes) Enabled

NETIF - Loopback Interface Options:

LWIP_NETIF_LOOPBACK (NETIF Loopback) Disabled

Infrastructure - Threading Options:

TCPIP_THREAD_NAME (TCPIP Thread Name) "tcpip_thread"
 TCPIP_THREAD_STACKSIZE (TCPIP Thread Stack Size) **2048 ***
 TCPIP_THREAD_PRIO (TCPIP Thread Priority Level) 24
 TCPIP_MBOX_SIZE (TCPIP Mailbox Size) 6

DEFAULT_THREAD_NAME (Default LwIP Thread Name)	"lwIP"
DEFAULT_THREAD_STACKSIZE (Default LwIP Thread Stack Size)	2048 *
DEFAULT_THREAD_PRIO (Default LwIP Thread Priority Level)	3
DEFAULT_RAW_RECVMBOX_SIZE (Default Mailbox Size on a NETCONN Raw)	0
DEFAULT_TCP_RECVMBOX_SIZE (Default Mailbox Size on a NETCONN TCP)	6
DEFAULT_ACCEPTMBOX_SIZE (Default Mailbox Size for Incoming Connections)	6

Thread Safe APIs - Netconn Options:

LWIP_NETCONN (NETCONN API)	Enabled
----------------------------	---------

Thread Safe APIs - Socket Options:

LWIP_SOCKET (Socket API)	Enabled
LWIP_COMPAT_SOCKETS (BSD-style Socket Functions Names)	1
LWIP_SOCKET_OFFSET (Socket Offset Number)	0
LWIP_SO_RCVTIMEO (Receive Timeout for Socket/Netconns)	Enabled *
LWIP_SOCKET_SELECT (Select for Socket)	Enabled
LWIP_SOCKET_POLL (Poll for Socket)	Enabled

3.16.3. PPP:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

PPP Options:

PPP_SUPPORT (PPP Module)	Disabled
--------------------------	----------

3.16.4. IPv6:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

IPv6 Options:

LWIP_IPV6 (IPv6 Protocol)	Disabled
---------------------------	----------

3.16.5. HTTPD:

Core(s) Settings:

Context(s): Cortex-M7
Initialized Context: Cortex-M7
Power Domain: D1

HTTPD Options:

LWIP_HTTPD (LwIP HTTPD Support ** CubeMX specific **) Disabled

3.16.6. SNMP:

Core(s) Settings:

Context(s): Cortex-M7
Initialized Context: Cortex-M7
Power Domain: D1

SNMP Options:

LWIP_SNMP (LwIP SNMP Agent) Disabled

3.16.7. SNTP/SMTP:

Core(s) Settings:

Context(s): Cortex-M7
Initialized Context: Cortex-M7
Power Domain: D1

SNTP Options:

LWIP_SNTP (LWIP SNTP Support ** CubeMX specific **) Disabled

SMTP Options:

LWIP_SMTP (LWIP SMTP Support ** CubeMX specific **) Disabled

3.16.8. MDNS/TFTP:

Core(s) Settings:

Context(s): Cortex-M7
Initialized Context: Cortex-M7
Power Domain: D1

MDNS Options:

LWIP_MDNS (Multicast DNS Support ** CubeMX specific **) Disabled

TFTP Options:

LWIP_TFTP (TFTP Support ** CubeMX specific **)

Disabled

3.16.9. Perf/Checks:

Core(s) Settings:

Context(s):

Cortex-M7

Initialized Context:

Cortex-M7

Power Domain:

D1

Sanity Checks:

LWIP_DISABLE_TCP_SANITY_CHECKS (TCP Sanity Checks)

Disabled

LWIP_DISABLE_MEMP_SANITY_CHECKS (MEMP Sanity Checks)

Disabled

Performance Options:

LWIP_PERF (Performace Testing for LwIP)

Disabled

3.16.10. Statistics:

Core(s) Settings:

Context(s):

Cortex-M7

Initialized Context:

Cortex-M7

Power Domain:

D1

Debug - Statistics Options:

LWIP_STATS (Statistic Collection)

Disabled

3.16.11. Checksum:

Core(s) Settings:

Context(s):

Cortex-M7

Initialized Context:

Cortex-M7

Power Domain:

D1

Infrastructure - Checksum Options:

CHECKSUM_BY_HARDWARE (Hardware Checksum ** CubeMX specific **)

Enabled

LWIP_CHECKSUM_CTRL_PER_NETIF (Generate/Check Checksum per Netif)

Disabled

CHECKSUM_GEN_IP (Generate Software Checksum for Outgoing IP Packets)

Disabled

CHECKSUM_GEN_UDP (Generate Software Checksum for Outgoing UDP Packets)

Disabled

CHECKSUM_GEN_TCP (Generate Software Checksum for Outgoing TCP Packets)

Disabled

CHECKSUM_GEN_ICMP (Generate Software Checksum for Outgoing ICMP Packets)

Enabled

CHECKSUM_GEN_ICMP6 (Generate Software Checksum for Outgoing ICMP6 Packets)	Disabled
CHECKSUM_CHECK_IP (Generate Software Checksum for Incoming IP Packets)	Disabled
CHECKSUM_CHECK_UDP (Generate Software Checksum for Incoming UDP Packets)	Disabled
CHECKSUM_CHECK_TCP (Generate Software Checksum for Incoming TCP Packets)	Disabled
CHECKSUM_CHECK_ICMP (Generate Software Checksum for Incoming ICMP Packets)	Enabled
CHECKSUM_CHECK_ICMP6 (Generate Software Checksum for Incoming ICMP6 Packets)	Disabled

3.16.12. Debug:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

LwIP Main Debugging Options:

LWIP_DBG_MIN_LEVEL (Minimum Level)	All
------------------------------------	-----

3.16.13. Platform Settings:

Driver_PHY	LAN8742
------------	---------

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
ETH	PC1	ETH_MDC	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PA1	ETH_REF_CLK	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PA2	ETH_MDIO	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PA7	ETH_CRS_DV	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PC4	ETH_RXD0	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PC5	ETH_RXD1	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PB13	ETH_TXD1	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PG11	ETH_TX_EN	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PG13	ETH_TXD0	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
I2C1	PB6	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D2
	PB7	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D2
I2C2	PB10	I2C2_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D2
	PB11	I2C2_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D2
RCC	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a		Cortex-M7* Cortex-M4	D3
SPI1	PA5	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PD7	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
TIM1	PE9	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PE11	TIM1_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
TIM3	PA6	TIM3_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PB5	TIM3_CH2	Alternate Function	No pull-up and no pull-	Low		Cortex-M4	D2

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
			Push Pull	down				
TIM4	PD12	TIM4_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
	PD13	TIM4_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
TIM8	PC7	TIM8_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M4	D2
Single Mapped Signals	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a			
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a			
	PH1-OSC_OUT (PH1)	RCC_OSC_OUT	n/a	n/a	n/a			
GPIO	PC13	GPIO_EXTI13	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PE8	GPIO_EXTI8	External Interrupt Mode with Falling edge	Pull-up *	n/a	VL_INT	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LD3	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PC6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	OLED_RST	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M1DIR	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M2DIR	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PD6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	BUZZ	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	OLED_CS	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PB9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	OLED_DC	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PE1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LD2	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4

* Initialized context

4.2. DMA configuration

DMA request	Stream	Direction	Priority
SPI1_TX	DMA2_Stream0	Memory To Peripheral	Medium *
TIM8_CH2	DMA1_Stream0	Memory To Peripheral	Low

SPI1_TX: DMA2_Stream0 DMA request Settings:

Mode: Normal
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

TIM8_CH2: DMA1_Stream0 DMA request Settings:

Mode: Normal
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Half Word
 Memory Data Width: Half Word

4.3. BDMA configuration

nothing configured in DMA service

4.4. MDMA configuration

nothing configured in DMA service

4.5. NVIC configuration

4.5.1. NVIC1

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
SPI1 global interrupt	true	5	0
TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts	true	15	0
DMA2 stream0 global interrupt	true	5	0
Ethernet global interrupt	true	6	0
Ethernet wake-up interrupt through EXTI line 86	true	5	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
CM4 send event interrupt for CM7	unused		
FPU global interrupt	unused		
HSEM1 global interrupt	unused		
Hold core interrupt	unused		

4.5.2. NVIC1 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
SPI1 global interrupt	false	true	true
TIM6 global interrupt, DAC1_CH1 and	false	true	true

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
DAC1_CH2 underrun error interrupts			
DMA2 stream0 global interrupt	false	true	true
Ethernet global interrupt	false	true	true
Ethernet wake-up interrupt through EXTI line 86	false	true	true

4.5.3. NVIC2

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
DMA1 stream0 global interrupt	true	5	0
EXTI line[9:5] interrupts	true	5	0
EXTI line[15:10] interrupts	true	5	0
TIM8 break interrupt and TIM12 global interrupt	true	5	0
TIM8 update interrupt and TIM13 global interrupt	true	5	0
TIM7 global interrupt	true	15	0
PVD and AVD interrupts through EXTI line 16		unused	
Flash global interrupt		unused	
TIM1 break interrupt		unused	
TIM1 update interrupt		unused	
TIM1 trigger and commutation interrupts		unused	
TIM1 capture compare interrupt		unused	
TIM3 global interrupt		unused	
TIM4 global interrupt		unused	
I2C1 event interrupt		unused	
I2C1 error interrupt		unused	
I2C2 event interrupt		unused	
I2C2 error interrupt		unused	
TIM8 trigger and commutation interrupts and TIM14 global interrupt		unused	
TIM8 capture compare interrupt		unused	
CM7 send event interrupt for CM4		unused	

Interrupt Table	Enable	Preenmption Priority	SubPriority
FPU global interrupt		unused	
HSEM2 global interrupt		unused	
Hold core interrupt		unused	

4.5.4. NVIC2 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
DMA1 stream0 global interrupt	false	true	true
EXTI line[9:5] interrupts	false	true	true
EXTI line[15:10] interrupts	false	true	true
TIM8 break interrupt and TIM12 global interrupt	false	true	true
TIM8 update interrupt and TIM13 global interrupt	false	true	true
TIM7 global interrupt	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Category viewContext Execution viewContext Initialization viewPower Domain view

Choose filters ...

by Context Execution

☐ Cortex-M7

☐ Cortex-M4

by Context Initialization

☐ Cortex-M7

☐ Cortex-M4

☒ None

by Power Domain

☐ D1

☐ D2

☐ D3

☒ None

Middleware

FREERTOS_M4

FREERTOS_M7

LWIP

Software Packs

CMSIS

5.1.2. Without filters

Category viewContext Execution viewContext Initialization viewPower Domain view

Choose filters ...

by Context Execution

☐ Cortex-M7☐ Cortex-M4

by Context Initialization

☐ Cortex-M7☐ Cortex-M4☒ None

by Power Domain

☐ D1☐ D2☐ D3☒ None

Middleware

FREERTOS_M4

FREERTOS_M7

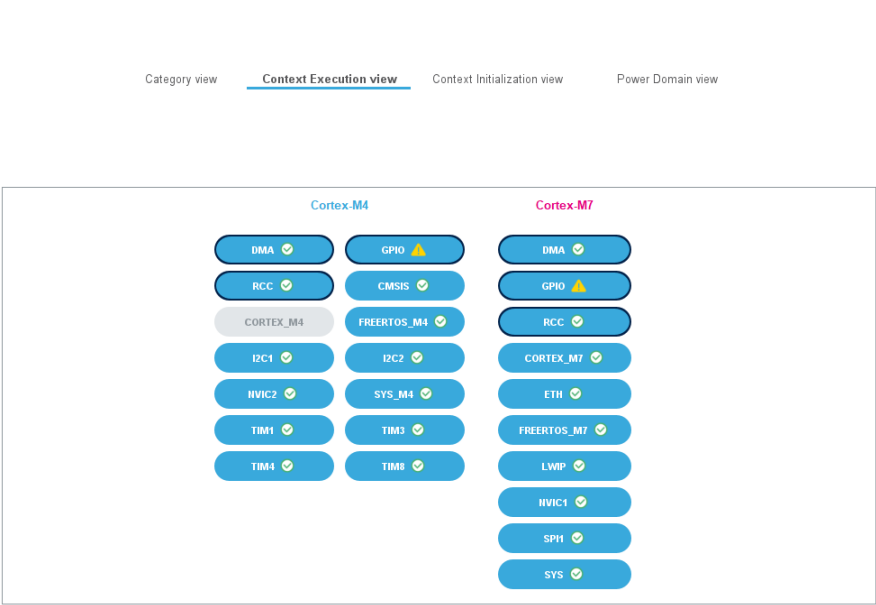
LWIP

Software Packs

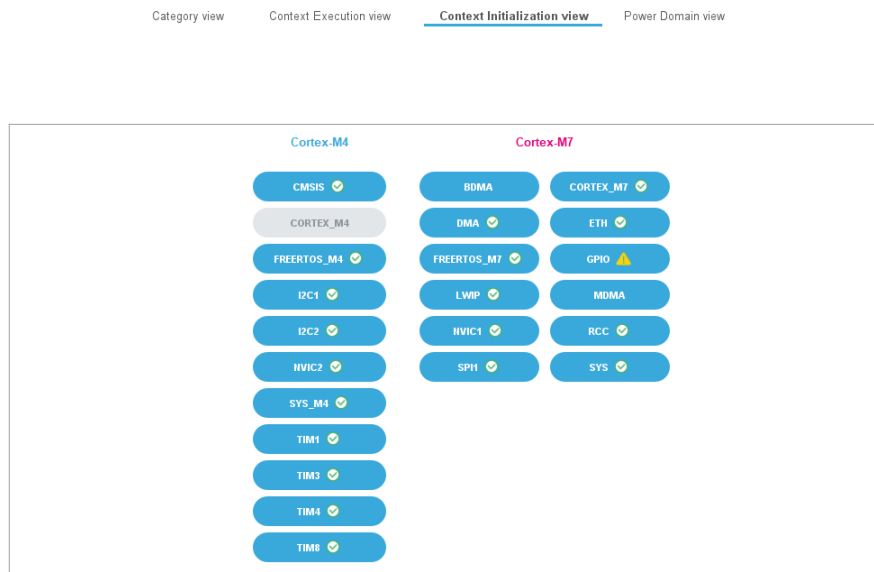
CMSIS

System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Utilities	Bsp	Other
BDMA		TIM1	ETH								
CORTEX_M4		TIM3	I2C1								
CORTEX_M7		TIM4	I2C2								
DMA		TIM8	SPH								
GPIO											
MDMA											
NVIC1											
NVIC2											
RCC											
SYS											
SYS_M4											

5.2. Context Execution view



5.3. Context Initialization view



5.4. Power Domain view



6. Software Pack Report

6.1. Software Pack selected

Vendor	Name	Version	Component
ARM	CMSIS	5.7.0	Class : CMSIS Group : DSP Variant : Library Version : 1.8.0

7. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7-bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32h7.pdf
Brochures	https://www.st.com/resource/en/brochure/humanoid-robot-reference-guide.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-

stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4990-getting-started-with-sigmadelta-digital-interface-on-applicable-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f4f7h7-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5286-stm32h7x5x7-dualcore-microcontroller-debugging-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5557-stm32h745755-and-stm32h747757-lines-dualcore-architecture-stmicroelectronics.pdf

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